

Predicting success with reduced dosing frequency of tralokinumab in patients with moderate-to-severe atopic dermatitis

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Abstract

Background Approved tralokinumab maintenance dosing regimens for treatment of moderate-to-severe atopic dermatitis (AD) include 300 mg every 2 weeks (Q2W) and every 4 weeks (Q4W). Clinicians may consider tralokinumab Q4W for patients whose skin has become clear or almost clear at week 16 with initial Q2W dosing.

Objectives To identify predictive factors associated with maintained response after switching to tralokinumab Q4W, evaluate recapture of treatment response after relapse on Q4W, and assess treatment-emergent immunogenicity with tralokinumab Q4W.

Methods These post hoc analyses utilized machine learning to identify predictive factors for maintained treatment response at week 52 using data from the week 16 responder population of the phase III ECZTRA 1 and 2 trials, i.e. patients who met Investigator's Global Assessment of clear/almost clear skin (IGA 0/1) and/or $\geq 75\%$ improvement in Eczema Area and Severity Index (EASI 75) at week 16 with tralokinumab Q2W monotherapy. Top-ranked factors were then assessed individually and together to identify factors associated with a similar maintained efficacy at week 52 between patients rerandomized to tralokinumab Q2W or Q4W monotherapy at week 16. Additionally, the probability of recapturing IGA 0/1 and/or EASI 75 response after relapse was assessed in patients on tralokinumab Q4W transferred to the open-label arm.

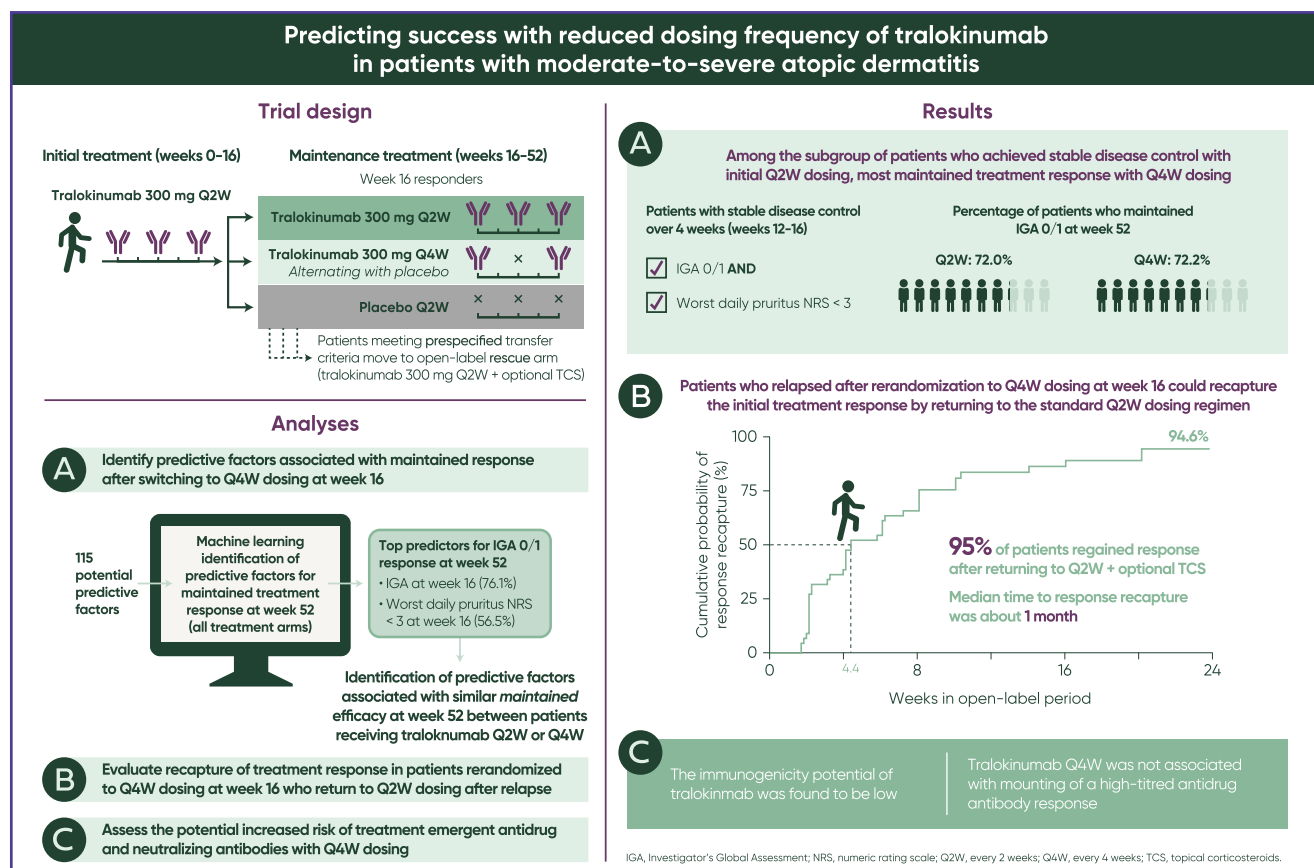
Results The two top-ranked predictive factors for maintained response at week 52 were IGA score at week 16 (76.1%) and worst daily pruritus numeric rating scale (NRS) < 3 at week 16 (56.5%). Patients whose AD reached stable clinical response scores of both IGA 0/1 and worst daily pruritus NRS < 3 from weeks 12–16 with tralokinumab Q2W similarly maintained IGA 0/1 response at week 52 regardless of dosing regimen beyond week 16 (72.0% of patients on Q2W and 72.2% of those on Q4W). Of patients who relapsed on Q4W, 94.6% recaptured treatment response after returning to Q2W dosing. The immunogenicity potential of tralokinumab was low, and patients with positive antidrug antibodies did not show loss of efficacy or higher incidences of adverse events.

Conclusions These data suggest that Q4W is an effective dosing regimen for most patients who achieved stable disease control as shown by clear/almost clear skin and no itch to mild itch over 4 consecutive weeks on the initial regimen of tralokinumab Q2W.

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Graphical Abstract



Lay summary

Atopic dermatitis (AD) is a chronic, itchy skin disease. Patients whose AD can't be controlled with topical (applied to the skin) prescription treatments often need long-term oral or injectable treatments. Tralokinumab is a biologic medication (derived from living organisms) that is injected under the skin. It is approved for treating AD at two different dosages: a standard initial dose of 300 mg every 2 weeks (Q2W) and a reduced dosing frequency option, every 4 weeks (Q4W) for patients who have achieved clear or almost clear skin with Q2W dosing.

In our study, we analysed factors associated with maintaining clear/almost clear skin after switching from tralokinumab Q2W to Q4W, using data from two clinical trials (ECZTRA 1 and 2) funded by LEO Pharma A/S. In these trials, patients initially received tralokinumab Q2W for 16 weeks. Those who responded well either continued with Q2W dosing for another 36 weeks or switched to less frequent Q4W dosing.

The top factors associated with maintenance of clear/almost clear skin were achievement of both clear/almost clear skin and experiencing no-to-mild itch symptoms at week 16. Among patients who experienced stable disease control with Q2W dosing (clear/almost clear skin and no-to-mild itch over a 4-week period), a similarly high percentage maintained clear/almost clear skin after switching to Q4W dosing as those who continued on Q2W dosing. Importantly, patients who worsened after switching to Q4W were able to regain disease control by returning to Q2W dosing.

Our findings can help identify patients who are most likely to successfully switch from tralokinumab Q2W to Q4W.

What is already known about this topic?

- Reducing the dosing frequency for biologics in treating psoriasis and rheumatoid arthritis has been shown to be a successful approach for patients with stable low disease activity or in clinical remission.
- Patients may prefer a reduced dosing frequency because of the convenience of less frequent injections and lowered yearly treatment cost.
- However, there are no clinical guidelines on dose tapering of biologics in patients with atopic dermatitis (AD).

What does this study add?

- Our study can help clinicians identify patients with moderate-to-severe AD who are eligible for dosing every 4 weeks (Q4W) with tralokinumab.
- Our data suggest that Q4W is an effective regimen for most patients who achieve stable disease control (clear/almost clear skin and no itch to mild itch symptoms over 4 consecutive weeks) on initial dosing every 2 weeks (Q2W).
- Patients who relapsed on Q4W recaptured the initial treatment response after returning to Q2W.

Atopic dermatitis (AD) is a chronic, relapsing inflammatory skin disease that typically requires long-term treatment.^{1,2} Biologics have recently become the first-line systemic treatment to prevent flares and achieve stable levels of minimal residual disease in patients with moderate-to-severe AD.^{3,4} Patients whose condition has improved on a biologic treatment may ask about reducing dosing frequency, because of the potential advantages, such as the increased convenience of less frequent injections and lowered yearly treatment costs.^{5,6} Although dose tapering of biologics in both psoriasis and rheumatoid arthritis has been shown to be an effective and safe approach for patients with stable low disease activity or in clinical remission,^{7,8} there are currently no clinical guidelines on dose tapering of biologics in patients with AD.

Tralokinumab is a fully human IgG4 high-affinity monoclonal antibody that specifically neutralizes interleukin (IL)-13,⁹ and which is currently approved in multiple countries for the treatment of moderate-to-severe AD.¹⁰ Approved tralokinumab dosing regimens include 300 mg every 2 weeks (Q2W), along with a reduced dosing frequency option, every 4 weeks (Q4W), which clinicians may consider for patients who achieved clear or almost clear skin at week 16 with the initial standard Q2W regimen.

In phase III clinical trials, tralokinumab Q2W with or without topical corticosteroids (TCS) resulted in significantly higher proportions of adults with moderate-to-severe AD who achieved the primary endpoints of Investigator's Global Assessment of clear or almost clear skin (IGA 0/1) and/or $\geq 75\%$ improvement in Eczema Area and Severity Index (EASI 75) at week 16 relative to placebo.^{11,12} Tralokinumab-treated patients who achieved IGA 0/1 and/or EASI 75 at week 16 were rerandomized to continue Q2W dosing, or switch to Q4W dosing or placebo. In the monotherapy trials, ECZTRA 1 and 2, 42% and 50% of patients rerandomized to tralokinumab Q4W dosing maintained IGA 0/1 or EASI 75 response, respectively, at week 52 without use of rescue medication, including any TCS use.¹³ In the TCS combination trial, ECZTRA 3, 78% and 91% of patients rerandomized to Q4W dosing maintained IGA 0/1 or EASI 75, respectively, at week 32.¹¹

Though these data are encouraging, the ability to predict which patients would be likely to maintain disease control with Q4W dosing could provide guidance on long-term dosing, and help reduce the number of patients who, having relapsed on Q4W, have to revert to Q2W dosing. There are currently no data available to help clinicians identify patients likely to maintain disease control with reduced dosing frequency, nor any data demonstrating regained AD disease control if treatment response is lost during less frequent dosing. Further, an association between less frequent dosing and an increased risk for the development of antidrug antibodies (ADA) has been reported for other biologic treatments.¹⁴

The objectives of these post hoc analyses were to (i) identify predictive factors associated with maintained response after switching to Q4W from Q2W; (ii) evaluate recapture of treatment response after a relapse on Q4W; and (iii) assess the potential increased risk of treatment emergent ADA and neutralizing antibodies (nAb) with tralokinumab Q4W.

Methods**ECZTRA 1 and 2 trial designs**

The design and methodology of ECZTRA 1 (NCT03131648) and 2 (NCT03160885) trials have been previously described.¹² Briefly, ECZTRA 1 and 2 were identically designed, multinational, double-blind, randomized, placebo-controlled, 52-week phase III trials of tralokinumab monotherapy in adults with moderate-to-severe AD (Figure S1; see [Supporting Information](#)). In the initial treatment period, patients were randomized 3 : 1 to receive either subcutaneous tralokinumab 300 mg (after an initial 600-mg loading dose on day 0) or placebo Q2W for 16 weeks. Tralokinumab-treated patients who achieved the prespecified primary endpoints (IGA 0/1 or EASI 75 at week 16) were rerandomized 2 : 2 : 1 to tralokinumab 300 mg monotherapy Q2W or Q4W or placebo for an additional 36-week maintenance treatment period. Patients who achieved the clinical response criteria with placebo continued to receive placebo Q2W to maintain blinding of the study. Patients who did not achieve IGA 0/1 or EASI 75 at week 16 were considered nonresponders and transferred to the open-label arm (tralokinumab Q2W plus optional TCS). During the blinded maintenance treatment period, patients were transferred to the open-label arm if they met the prespecified transfer criteria listed in Figure 1. Patients who were transferred to the open-label arm continued their scheduled visit sequence.

Analyses

All analyses in the manuscript are post hoc and based on the week 16 responder population (i.e. patients who achieved IGA 0/1 and/or EASI 75 at week 16 with tralokinumab Q2W monotherapy) using pooled data from the ECZTRA 1 and 2 studies unless otherwise specified.

Identifying factors associated with successful tralokinumab dosing every 4 weeks

Machine learning was used to identify predictive factors for ECZTRA 1 and 2 tralokinumab Q2W week 16 responders (IGA 0/1 and/or EASI 75 without rescue medication) to maintain response at week 52, using 115 factors (63 main

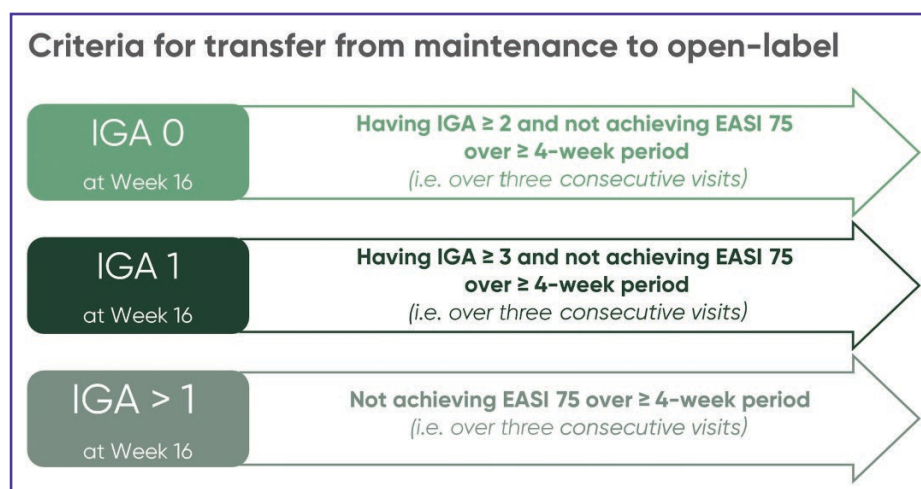


Figure 1 Predefined criteria for transfer from blinded maintenance treatment to the open-label arm in ECZTRA 1 and 2. EASI 75, improvement of at least 75% in Eczema Area and Severity Index; IGA, Investigator's Global Assessment.

factors including, but not limited to, baseline characteristics, medical history, efficacy after 16 weeks or earlier, randomized treatment and 52 interactions). The algorithm utilized 1000 replications. Per-replication stepwise selection based on logistic regression was used to systematically search combinations of significant ($P < 0.05$) predictive factors for maintenance of IGA 0/1 without rescue use at week 52. Data were randomly split into training (75% of data) and validation (25%) datasets. These predictive factors were ranked by percentage of times identified by the algorithm in the model for IGA 0/1 at week 52, where a higher value indicated greater potential predictive ability.

To identify factors associated with a similar maintained efficacy at week 52 between responders rerandomized to tralokinumab Q2W and Q4W monotherapy at week 16, top-ranked factors were assessed individually and together in subgroup analyses. These predictive factors were also explored using data from the continuation period of the concomitant TCS trial, ECZTRA 3 (NCT03363854). The design and methodology of this trial has been previously described.¹¹

Evaluating recapture of treatment response after a relapse

Among week 16 responders rerandomized to tralokinumab Q4W, the probability of recapturing IGA 0/1 and/or EASI 75 response after transfer to the open-label arm was assessed using Kaplan–Meier estimates. Time was set to zero upon transfer to the open-label arm.

Evaluation of immunogenicity with tralokinumab dosing every 4 weeks

All clinical trials with tralokinumab included testing of serum samples for ADA and, if positive, also for nAb. Blood samples were collected at prespecified timepoints during treatment periods and at safety follow-up visits. ADA and nAb responses to tralokinumab were measured using semi-homogeneous bridging and binding electrochemiluminescence assays, respectively, combined with acid dissociation

and magnetic bead extraction to enhance drug tolerance and sensitivity. The ADA assay utilized biotinylated tralokinumab bound to streptavidin-coated magnetic beads for capture of ADA and detection by sulfo-tagged tralokinumab, and addition of read buffer containing tripolyamine. The nAb assay utilized biotinylated tralokinumab on streptavidin-coated magnetic beads for capture of the neutralizing antibodies and detection using sulfo-tagged IL-13. Thereby, the chemiluminescent signal was indirectly proportional to the concentration of neutralizing anti-tralokinumab antibodies.

The frequency of patients testing positive for ADA was assessed during the maintenance treatment period (at weeks 28 and 52, and at the safety follow-up visit at week 66) for week 16 responders rerandomized to tralokinumab Q2W or Q4W at week 16. To explore whether switching back to Q2W after reducing the dosing frequency to Q4W impacts the ADA response, all patients entering the open-label extension trial, ECZTEND (NCT03587805), from either Q2W or Q4W maintenance treatment in ECZTRA 1 or 2, were assessed for ADA in an interim analysis using a data cut-off of 31 December 2021. Patients had been enrolled for at least a year in ECZTEND at data cut-off. The design and methodology of the ECZTEND trial has been previously described.¹⁵

To identify adverse events (AEs) with special relevance to immunogenicity, standardized MedDRA searches for anaphylaxis and serious allergic reactions and immune complex disease were conducted (MedDRA searches are listed in Tables S1–S3; see [Supporting Information](#)).

Results

Patient disposition and baseline characteristics

Patients ($N=337$) initially treated with tralokinumab Q2W who achieved IGA 0/1 and/or EASI 75 at week 16 without any systemic or topical rescue treatment in the ECZTRA 1 and 2 trials were included in the current analyses. Baseline demographics and clinical characteristics were generally well balanced across treatment groups (Table 1).

Table 1 Baseline demographics and clinical characteristics for pooled ECZTRA 1 and 2 week 16 responders grouped by week 16 rerandomization

	Rerandomization at week 16		
	Tralokinumab Q2W (N= 130)	Tralokinumab Q4W (N= 134)	Placebo Q2W (N= 73)
Age (years), mean (SD)	37.1 (13.3)	38.9 (16.3)	38.3 (13.2)
Male, <i>n</i> (%)	74 (56.9)	68 (50.7)	41 (56)
BSA involvement (%), mean (SD)	44.1 (22.1)	44.8 (22.2)	45.3 (25.0)
Duration of AD (years), mean (SD)	27.6 (15.6)	27.1 (15.3)	30.4 (15.4)
IGA, <i>n</i> (%)			
IGA 3 (moderate)	94 (72.3)	75 (56.0)	45 (62)
IGA 4 (severe)	36 (27.7)	59 (44.0)	28 (38)
EASI, mean (SD)	27.6 (11.7)	27.7 (11.5)	29.0 (13)
DLQI, mean (SD) ^a	16.0 (7.6), N= 128	15.2 (7.0), N= 132	16.6 (6), N= 72
Weekly average worst daily pruritus	7.5 (1.5),	7.6 (1.5),	7.7 (1),
NRS, mean (SD) ^a	N= 129	N= 133	N= 73

AD, atopic dermatitis; BSA, body surface area; DLQI, Dermatology Life Quality Index; EASI, Eczema Area and Severity Index; IGA, Investigator's Global Assessment; *N*, total number of patients per group; *n*, number of patients meeting indicated metric; NRS, numeric rating scale; Q2W, every 2 weeks; Q4W, every 4 weeks. ^aIndividual row *N* values are shown when different from the respective total number of patients in the group.

Factors associated with maintained tralokinumab treatment response at week 52

In ECZTRA 1 and 2, among week 16 responders who were rerandomized to tralokinumab Q2W, tralokinumab Q4W or placebo, IGA 0/1 or EASI 75 was maintained without any systemic or topical rescue use at week 52 in 56.2% (73 of 130) ($P<0.001$), 50.0% (67 of 134) ($P=0.003$) and 27% of patients (20 of 73), respectively. Utilizing these data, the machine learning algorithm performed 1000 replications and identified IGA score (the lower the better) at week 16 (76.1%) and worst daily pruritus numeric rating scale (NRS) <3 at week 16 (56.5%) as the top-ranked predictive factors for maintained response (IGA 0/1) at week 52 with tralokinumab monotherapy (Table 2).

Factors associated with successful tralokinumab dosing every 4 weeks

In patients who achieved both IGA 0/1 and worst daily pruritus NRS <3 at week 16, the proportion of patients who maintained response at week 52 was greater for both Q2W and Q4W than for patients only achieving IGA 0/1. However, a higher proportion of patients maintained response with Q2W relative to Q4W at week 52 (Figure 2).

Table 2 Top three predictive factors for maintenance of Investigator's Global Assessment of clear/almost clear skin (IGA 0/1) at week 52 with tralokinumab monotherapy among week 16 responders in ECZTRA 1 and 2

Predictive factor of IGA 0/1 at week 52	% Identified ^a
IGA score (the lower the better) at week 16	76.1
Worst daily pruritus NRS <3 at week 16 (Yes/No)	56.5
EASI 90 at week 16 (Yes/No)	35.4

EASI 90, improvement of at least 90% in Eczema Area and Severity Index score; NRS, numeric rating scale. ^aFactors ranked by percentage of 1000 replications in which each was identified as a significant ($P<0.05$) predictive factor for IGA 0/1 without rescue use at week 52 based on logistic regression. Use of rescue medication (including topical corticosteroids) considered as nonresponse and missing data imputed as non-response; 1000 replications.

As AD is a fluctuating disease, we assessed the impact of the stable disease component in patients who met both IGA 0/1 and worst daily pruritus NRS <3 over three consecutive visits (weeks 12, 14 and 16). Achievement of both IGA 0/1 and worst daily pruritus NRS <3 from weeks 12–16, compared with week 16 alone, resulted in a substantial increase in the proportion of patients who maintained IGA 0/1 response at week 52 with Q4W (72.2%), which was comparable with the result achieved with continued Q2W (72.0%) (Figure 3).

Achievement of both IGA 0/1 and worst daily pruritus NRS <3 at week 12 alone was associated with differing week 52 response rates between dosing regimens (IGA 0/1 at week 52 was 71.9% for Q2W and 54.8% for Q4W) (Figure 4). These results suggest that achievement of a stable response over time is a better predictor than an early response.

Lastly, these findings from the pooled monotherapy trials data were explored using data from the concomitant TCS trial, ECZTRA 3. In this analysis, a high proportion (90.0%) of patients who achieved stable IGA 0/1 and worst daily pruritus NRS <3 at weeks 12–16 with tralokinumab Q2W plus TCS maintained IGA 0/1 at week 32 after being rerandomized to tralokinumab Q4W plus TCS at week 16, a proportion comparable to the tralokinumab Q2W plus TCS dosing regimen (94.1%) (Figure S2; see [Supporting Information](#)).

Recapture of treatment response in patients relapsing on tralokinumab dosing every 4 weeks

The median time to response recapture with tralokinumab Q2W plus optional TCS was 4.4 weeks (31 days) for patients relapsing after extending the tralokinumab dosing interval to Q4W. Nearly all patients (94.6%) recaptured IGA 0/1 or EASI 75 response at 20 weeks of retreatment with tralokinumab Q2W plus optional TCS (Figure 5).

Optional TCS use was reported by 23.5% of patients who transferred to the open-label arm from the tralokinumab Q4W group, none of which was high-potency TCS.

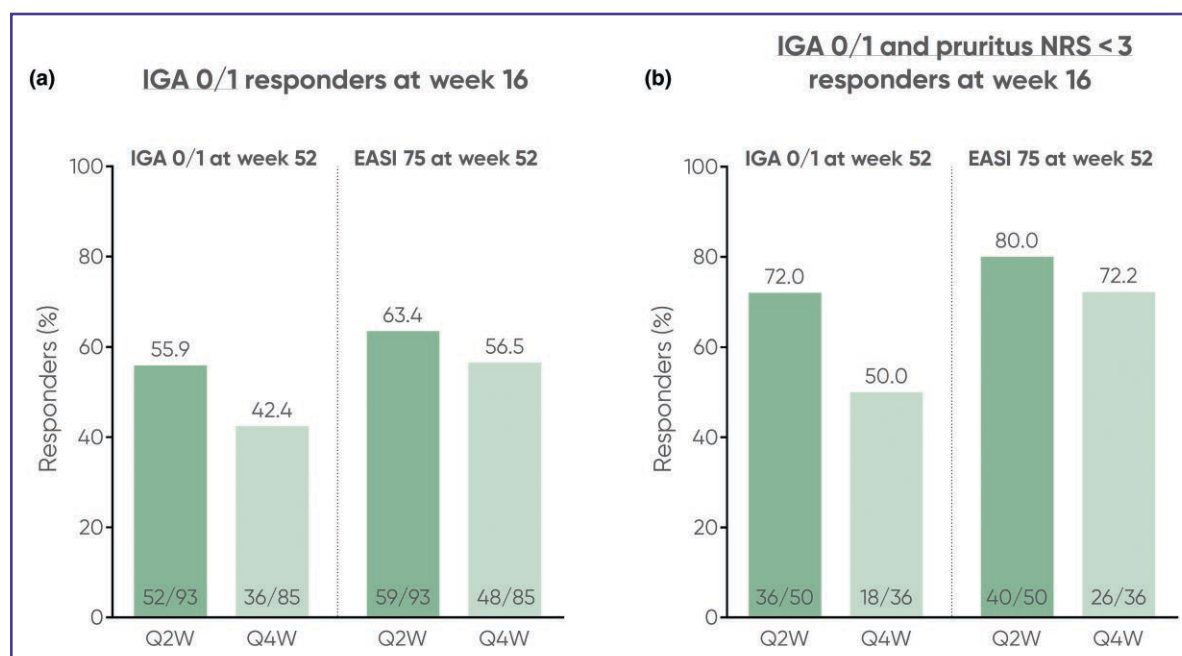


Figure 2 Maintenance of tralokinumab treatment response at week 52 in week 16 responders who achieved (a) Investigator's Global Assessment of clear/almost clear skin (IGA 0/1) at week 16 and (b) IGA 0/1 and worst daily pruritus numeric rating scale (NRS) score <3 at week 16 in ECZTRA 1 and 2. Number of patients meeting the indicated metric/total number of patients per group (*n/N*) is listed within the bottom of each respective bar. EASI 75, improvement of at least 75% in Eczema Area and Severity Index; Q2W, every 2 weeks; Q4W, every 4 weeks.

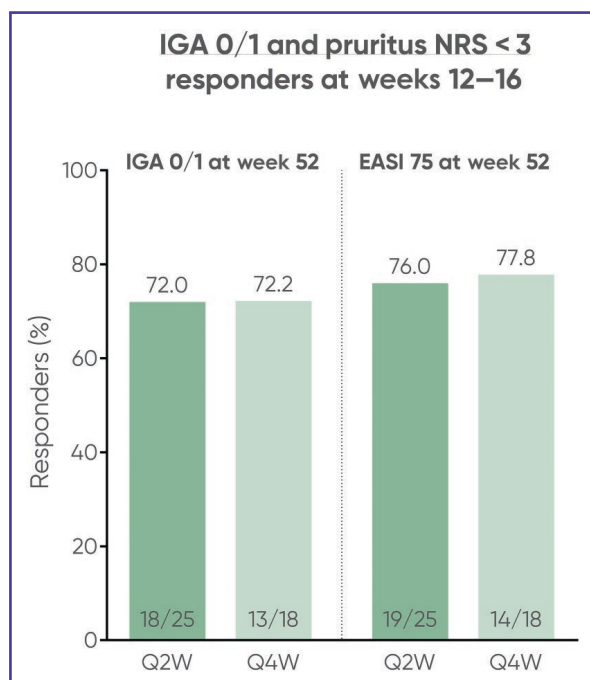


Figure 3 Maintenance of tralokinumab treatment response at week 52 in week 16 responders who achieved Investigator's Global Assessment of clear/almost clear skin (IGA 0/1) and worst daily pruritus numeric rating scale (NRS) score <3 over weeks 12–16 in ECZTRA 1 and 2. Number of patients meeting the indicated metric/total number of patients per group (*n/N*) is listed within the bottom of each respective bar. EASI 75, improvement of at least 75% in Eczema Area and Severity Index; Q2W, every 2 weeks; Q4W, every 4 weeks.

Evaluation of immunogenicity with tralokinumab dosing every 4 weeks

Frequencies for ADA and nAb during the initial 16-week treatment period with tralokinumab Q2W in the ECZTRA 1, 2, 3 and 5 trials have been previously reported.¹⁶ Frequencies of ADA and nAb were low and similar for patients receiving tralokinumab [ADA: 1.4% (21 of 1553); nAb: 0.1% (2 of 1553)] and placebo [ADA: 1.3% (8 of 629); nAb: 0.2% (1 of 629)]. All treatment-emergent ADA were low titre (ranging from <10 to 640), with no increased risk of anaphylaxis, serious allergic reactions, immune-complex disease, serum sickness or serum sickness-like reactions.¹⁶

During the blinded maintenance period of the ECZTRA 1 and 2 trials, a lower ADA incidence was observed in the tralokinumab Q2W group (1.3%; 2 of 159) compared with the Q4W group (6.1%; 10 of 165). However, an additional 4.1% of patients treated with tralokinumab Q2W plus optional TCS in the open-label treatment arm had a positive ADA response. ADA titres were generally low (<10 to 40); 7 of the 10 patients in the Q4W treatment group had ADA titres ≤10, and 2 patients were positive for nAb (Table 3). Additionally, eight of the 10 patients who were ADA positive and on tralokinumab Q4W presented only one positive ADA sample during the trial. All IGA scores in patients who were ADA positive were clear, almost clear or mild, and EASI ranged from 0.6 to 11.2 (Table 3). Specifically, for the two patients with positive nAb results, IGA scores were almost clear and mild while EASI scores were 1.3 and 2.3, respectively. No AEs of anaphylaxis/serious allergic reactions or immune complex disease were observed in patients

Table 3 Week 16 responders with positive antidrug antibody (ADA) status during maintenance treatment period or safety follow-up in ECZTRA 1 and 2

Dose	Visit with ADA	ADA ^a	nAb	Tralokinumab concentration (µg mL ⁻¹)	IGA at visit with ADA	EASI at visit with ADA
Q2W	Week 52	40	Negative	90.0	1	11.2
Q2W	Week 52	20	Negative	121	0	0.6
Q4W	Week 52	40	Negative	20.4	2	2.3
Q4W	Safety FU	10	Negative	2.97	NA	NA
Q4W	Week 52	< 10	Negative	25.3	2	2.8
Q4W	Safety FU	< 10	Negative	3.64	NA	NA
Q4W	Week 28	10	Negative	38.4	2	5.6
Q4W	Week 28	< 10	Negative	66.5	1	2.2
Q4W	Week 30	< 10	Negative	79.7	1	4.0
Q4W	Week 52	< 10	Positive	61.3	1	1.3
Q4W	Safety FU	< 10	Negative	5.34	NA	NA
Q4W	Week 28	< 10	Positive	72.3	2	2.3
Q4W	Week 50	< 10	Negative	63.5	2	3.6

EASI, Eczema Area and Severity Index; FU, follow-up; IGA, Investigator's Global Assessment; NA, not applicable; nAb, neutralizing antibodies; Q2W, every 2 weeks; Q4W, every 4 weeks. ^aMinimum required dilution was 10.

who were ADA positive during the maintenance treatment period.

Further analyses were conducted to explore whether switching back to Q2W after reducing dose frequency to Q4W impacted the ADA response. Two hundred and seven patients from the blinded Q2W and Q4W groups in ECZTRA 1 and 2 were subsequently enrolled in the open-label ECZTEND trial, receiving tralokinumab Q2W, for up to 3 years; none of these patients had a positive ADA response as of the cut-off date.

Discussion

The present post hoc analyses provide insights that should help inform clinicians on whether to downtitrate tralokinumab to Q4W dosing in clinical practice. Utilizing a machine learning approach, we identified IGA score (the lower the better) at week 16 and worst daily pruritus NRS <3 at week 16 with tralokinumab Q2W monotherapy as the top-ranked predictive factors for maintained IGA 0/1 response at week 52, regardless of which treatment regimen patients were rerandomized to in

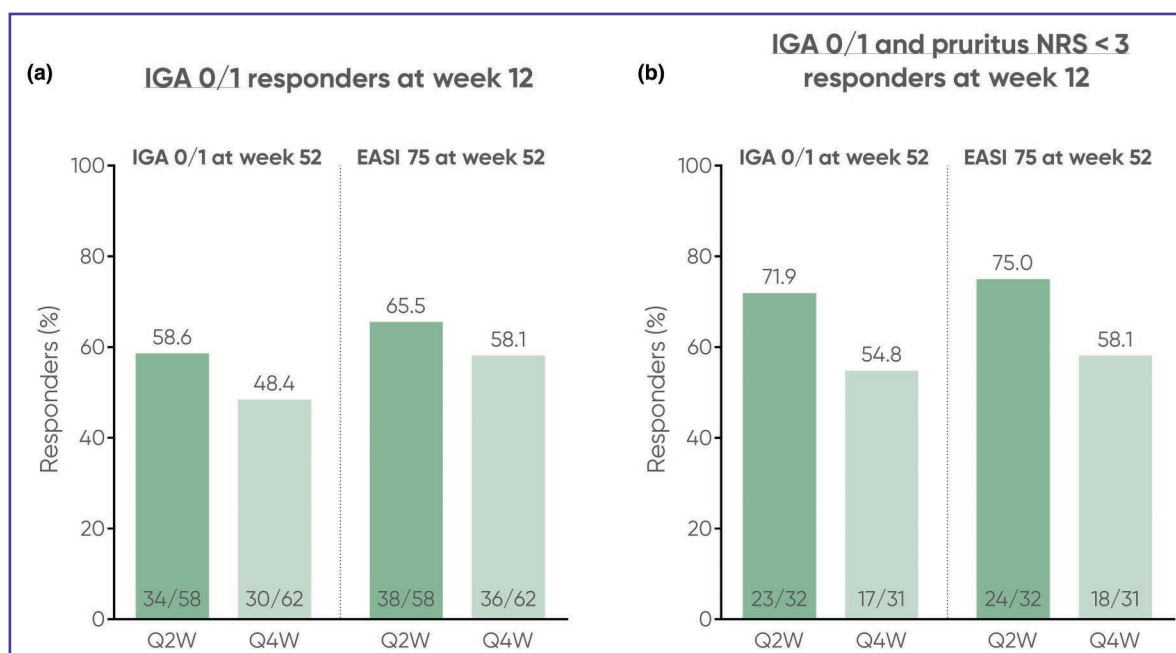


Figure 4 Maintenance of tralokinumab treatment response at week 52 in week 16 responders who achieved (a) Investigator's Global Assessment of clear/almost clear skin (IGA 0/1) at week 12 and (b) IGA 0/1 and worst daily pruritus numeric rating scale (NRS) score <3 at week 12 in ECZTRA 1 and 2. Number of patients meeting the indicated metric/total number of patients per group (n/N) is listed within the bottom of each respective bar. EASI 75, improvement of at least 75% in Eczema Area and Severity Index; Q2W, every 2 weeks; Q4W, every 4 weeks.

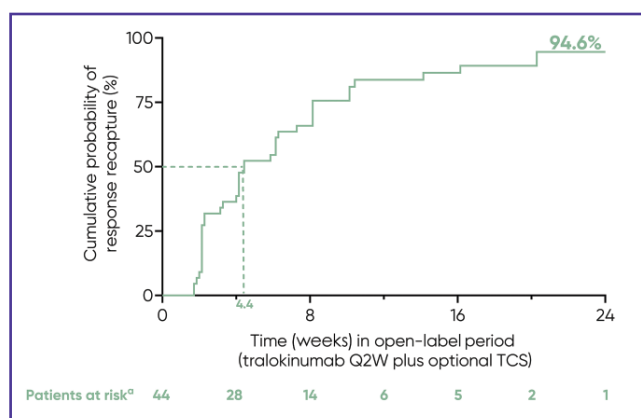


Figure 5 Time to recapture Investigator's Global Assessment of clear/almost clear skin (IGA 0/1) and/or improvement of at least 75% in Eczema Area and Severity Index (EASI 75) in week 16 responders rerandomized to every 4 weeks (Q4W) and subsequently transferred to the open-label arm [tralokinumab every two weeks (Q2W) plus optional topical corticosteroids (TCS)] in ECZTRA 1 and 2. Transfer criteria: week 16 IGA 0/1 responders, at least a 2-point deterioration in IGA and loss of EASI 75 response over at least 4 weeks; week 16 EASI 75 responders, loss of EASI 75 response over at least 4 weeks. Week is relative to week of transfer to open-label arm. In the open-label arm, patients only attended the clinic every 4 weeks (home-dosing). Graph truncated at week 24 due to very few patients having > 24 weeks of treatment in the open-label arm. ^aPatients who did not previously recapture response and are still in the trial.

the maintenance phase. These factors were then used to further explore combinations that could achieve similar maintained efficacy between Q4W and Q2W dosing options at week 52.

Achieving the top-ranked predictive factors at week 16, individually or in combination, increased the proportion of IGA 0/1 responders at week 52 for both dosing options. However, similar maintained efficacy between the Q4W and Q2W dosing options at week 52 was only shown when both of these factors were achieved over 4 consecutive weeks (weeks 12–16). In this scenario, over 70% of patients maintained IGA 0/1 at week 52 with both dosing options, without using rescue medication (including TCS). Using the same response criteria and data from the ECZTRA 3 trial, which allowed concomitant TCS as needed on active lesions, the proportion of patients who maintained IGA 0/1 response increased to over 90% at week 32 for both dosing options. These results suggest that a stable, deeper skin clearance response like IGA 0/1 at week 16, combined with robust itch relief, may help clinicians identify patients with a high likelihood of long-term success with tralokinumab Q4W.

Our results align with the eligibility criteria commonly used for dose tapering in studies of biologics in psoriasis, reflecting a state of stable clinical remission or low disease activity.⁷ Previous work has shown that IL-13 levels in lesional and nonlesional skin correlate with the extent and severity of lesions and itch measured by SCORAD (SCORing Atopic Dermatitis), consistent with the role of IL-13 as a key driver of the underlying inflammation in AD.^{17,18} Thus, following a period of clear/almost clear skin and low levels of itch achieved with tralokinumab Q2W, it is possible that IL-13-mediated inflammation in the skin is attenuated, and less frequent dosing of the antibody is required to neutralize the proinflammatory effects of the cytokine.

Previous work examining varying dosing regimens of dupilumab showed a reduction in treatment response (including EASI and pruritus NRS) with less frequent dosing regimens (Q4W or Q8W).¹⁹ More recently, real-world studies have found that achievement of stable low disease activity (EASI ≤ 7) with dupilumab Q2W dosing for ≥ 6 months was generally associated with maintained treatment response with less frequent (Q3W or Q4W) dosing.^{20,21}

Importantly, our analyses also suggest that most patients who relapse after extending the tralokinumab dosing interval to Q4W can recapture the treatment response by switching back to the Q2W dosing regimen. The median time to response recapture was 31 days and about 95% of patients had recaptured the treatment response within 20 weeks. Although the optional TCS use allowed in the open-label arm may have contributed to regained response, < 25% of patients reported any TCS use during the open-label treatment period, of which none was classified as high potency TCS.

Lastly, therapeutic monoclonal antibodies have the potential to induce an immunological response leading to generation of ADA and nAb.¹⁴ It is of particular concern if patients become immune-sensitized during less frequent Q4W dosing, as they may be at risk of an increased ADA response when switching back to Q2W. Potential clinical consequences of immunogenicity known from other monoclonal antibodies include reduced efficacy and/or increased safety concerns, such as induction of anaphylaxis/hypersensitivity reactions or immune-complex disease.²² Our analysis of available immunogenicity data, including patients treated with Q4W dosing during the maintenance phase in the ECZTRA 1 and 2 trials who were switched to Q2W dosing upon enrolment in the long-term extension trial, ECZTEND, suggests that the immunogenicity potential of tralokinumab is low, any presence of ADA is transient and that, overall, tralokinumab Q4W dosing is not associated with mounting of a high-titre ADA response. There was no indication of an overall impact of ADA with tralokinumab on efficacy or safety. In addition, although tralokinumab Q2W has an excellent long-term safety profile,¹⁵ a slightly lower AE frequency has been reported previously for the Q4W dosing option compared with Q2W.¹²

Limitations of these analyses include their post hoc nature and the relatively low number of patients. In addition, the analyses to identify potential predictors, due to limitations in the clinical study designs, utilized patients who could only be switched to the Q4W option at week 16. Recent data have shown that week 16 may be too early for evaluating the full benefit of tralokinumab and that continued treatment with tralokinumab Q2W after week 16 is associated with further improvement.^{13,23} Therefore, patients with progressive improvements who may have become eligible for Q4W dosing after week 16 could not be accounted for in our analyses. However, a strength of the recapture response analyses is that the decision to switch patients back to Q2W dosing was based on pre-specified criteria that mirror what would be considered a worsening of the disease that requires treatment change in clinical practice.

The current analyses suggest that tralokinumab Q4W is an effective dosing regimen for most patients who achieved stable disease control – as shown by clear/almost clear skin

and no-to-mild itch over 4 consecutive weeks – with the standard Q2W dosing regimen. Moreover, patients who relapse after switching to tralokinumab Q4W can recapture treatment response by returning to Q2W. These data can help clinicians assess the optimal tralokinumab maintenance dosing frequency for patients with AD over time.

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Conflicts of interest

S.W. is a speaker, advisory board member and/or investigator for AbbVie, Ammirall, Boehringer Ingelheim, Eli Lilly, Galderma, GSK, LEO Pharma, Pfizer, Regeneron and Sanofi; he has also received institutional research grants from LEO Pharma, Pfizer and Sanofi. A.B. has served as a consultant to AbbVie, Ammirall, Celgene, Galderma, Janssen, LEO Pharma, Lilly, Novartis, Sanofi and UCB; he has participated in advisory boards for The Psoriasis Association, Changing Faces, Ichthyosis Support Group and National Eczema Association; he has received grants from the European Academy of Dermatology and Venereology, and travel grants from Ammirall, Janssen and LEO Pharma; he has participated in guidelines committees of the British Association of Dermatologists; and he is working as an Editor for the book *Practical Psychodermatology*, as president for the European Society for Dermatology and Psychiatry, and as a chair for an All Party Parliamentary Group on Skin. H.C.-h.H. is a researcher, consultant and/or advisor for AbbVie, Amgen, Arcutis, ASLAN Pharmaceuticals, Bausch Health, Boehringer Ingelheim, Bristol Myers Squibb, Celgene, Dermavant, Dermira, DS Biopharma, Eli Lilly, Evelo, Galderma, GSK, Incyte, Janssen, LEO Pharma, MedImmune, Novartis, Pfizer, Regeneron, Roche, Sanofi Genzyme, Sun Pharma and UCB. J.F.S. has served as a speaker, advisory board member and/or investigator for AbbVie, Ammirall, Amgen, Astra Zeneca, Bristol Myers Squibb, Eli Lilly, Galderma, Incyte, LEO Pharma, Novartis, Pfizer, Regeneron and Sanofi. K.P. reports grants or personal fees for participation in advisory boards from AbbVie, Ammirall, Galderma, Janssen, LEO Pharma, Lilly, Novartis, Pierre Fabre, Sanofi and Sun Pharma. A.W. has received grants, personal fees or nonfinancial support from AbbVie, Aileens Pharma, Alentis Therapeutics, Ammirall, Amgen, Beiersdorf, Bioderma, Bristol Myers Squibb, Chugai Pharmaceutical Co., Galapagos Pharmaceuticals, Galderma, GSK, Hans Karrer, Janssen, LEO Pharma, Lilly, L'Oréal, Maruho, MedImmune, Merck, MSD, Novartis, Pfizer, Pierre Fabre, Regeneron, Sandoz, Santen Pharmaceutical

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Data availability

The data underlying this article will be made available on request to the study sponsor, following review by the external Patient and Scientific Review Board.

Ethics statement

The ECZTRA 1, 2, 3 and ECZTEND trials were sponsored by LEO Pharma A/S (Ballerup, Denmark) and conducted in accordance with the ethical principles derived from international guidelines, including the Declaration of Helsinki and Council for International Organizations of Medical Sciences International Ethical Guidelines, and in compliance with ICH guidelines for Good Clinical Practice. The clinical trial was approved by institutional review boards or ethics committees at each study site. This trial followed the CONSORT reporting guideline.

Patient consent

Not applicable.

Supporting Information

Additional [Supporting Information](#) may be found in the online version of this article at the publisher's website.

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